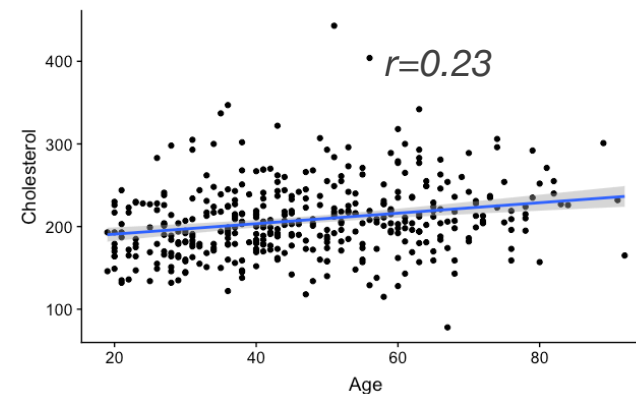
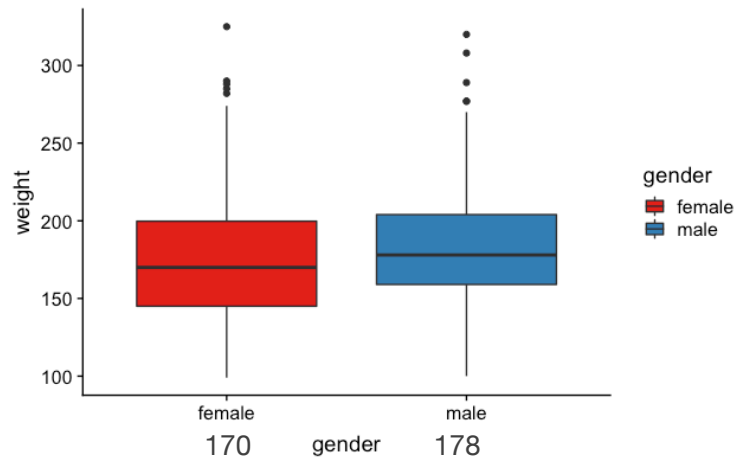


7. Hypothesis testing

7.1 Principle

Are observations significant?



- For the cohort, we observe:
 - a difference in man/women weights
 - a non-zero correlation between age and cholesterol
- But:
 - would we observe this in another cohort??
 - Does this hold for the entire (unknown) population?
→ *is this difference/correlation significant?*

Hypothesis testing: what do we need?

- **Question** that we want to investigate:
 - *is there a **GENERAL** weight difference between men/women?*
 - *is there a **GENERAL** non-zero correlation between age/cholesterol?*
- **Hypothesis:**
 - **Null-hypothesis (H_0):** this is the “no-effect” Hypothesis
 - ▶ no difference between the **expectations of the random variables** $E(X_m) = E(X_w)$
 X_m =weight of Men and X_w = weights of Women
 - ▶ no correlation between the random variables X_{chol} and X_{age} : $cor(X_{age}, X_{chol}) = 0$
 - **Alternative hypothesis (H_1):**
 - ▶ $E(X_m) \neq E(X_w)$
 - ▶ $cor(X_{age}, X_{chol}) \neq 0$
- **Test-statistics:** numerical value that can be computed from the data, with known distribution under H_0

Example

- **Study:** effect of fertilizer F1 on plant growth
 - no-fertilizer: $h = 1.5m$
 - fertilizer on $n = 10$ samples:
 $= \{1.47, 1.462, 1.51, 1.61, 1.27, 1.51, 1.55, 1.49, 1.44, 1.5\}$
- **Random variable:** plant height X after treatment with F1
- **Question:** *does the treatment with fertilizer **enhance** plant growth?*

- **Hypothesis:**

- H_0 : no

$$E(X) \leq h = 1.5m$$

- H_1 : yes

$$E(X) > h = 1.5m$$

- Effect size:
- size of random effect:

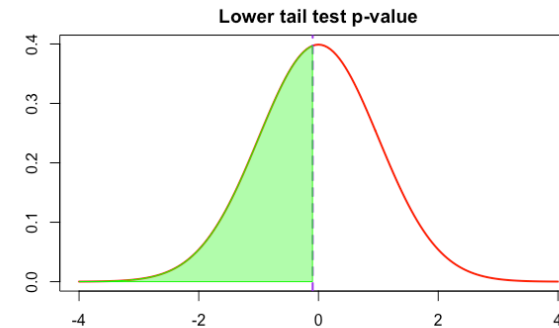
$$\left. \begin{array}{l} \bar{x} - h = -0.003 \\ s/\sqrt{n} = 0.031 \end{array} \right\} t = \frac{\bar{x} - h}{s/\sqrt{n}} = -0.09$$

s = standard
deviation
of sample

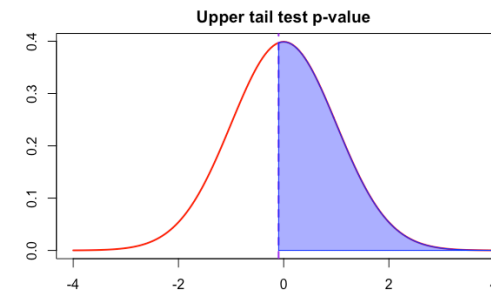
- What are typical values of t **under the H_0 hypothesis?**

Example

- Distribution of t under the H_0 hypothesis
- Vertical line = observed value of test statistics t
- Green = probability to observe under H_0 a lower value of t
- Blue = probability to observe under H_0 a larger value of t
- Here: Blue = 53.9% of total area



Pvalue = 0.461



Pvalue = 0.539

Conclusion: if H_0 (= no effect) is true, there is a **53.9% probability** to observe a value of t larger or equal to the one observed

→ **not unlikely, hence no reason to distrust H_0 (= no effect)**

Example

- **Study:** effect of fertilizer F2 on plant growth

- no-fertilizer: $h = 1.5m$
- fertilizer on $n = 10$ samples:

$$X = \{1.47, 1.62, 1.61, 1.61, 1.47, 1.51, 1.55, 1.59, 1.64, 1.5\}$$

- **Random variable:** plant height X after treatment with F2
- **Question:** *does the treatment with fertilizer **enhance** plant growth?*
- **Hypothesis:**

- H_0 : no

$$E(X) \leq h = 1.5m$$

- H_1 : yes

$$E(X) > h = 1.5m$$

- Effect size:
- size of random effect:

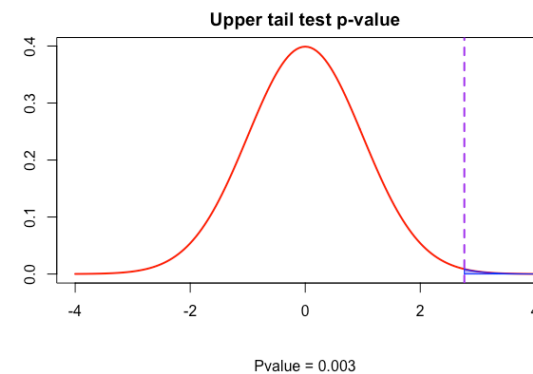
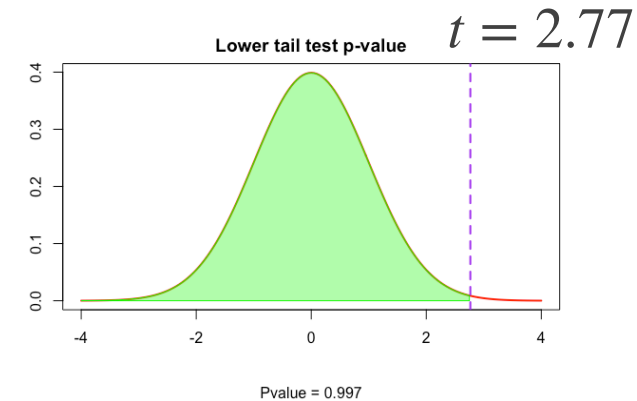
$$\left. \begin{array}{l} \bar{x} - h = 0.057 \\ s/\sqrt{n} = 0.02 \end{array} \right\} t = \frac{\bar{x} - h}{s/\sqrt{n}} = 2.77$$

s = standard
deviation
of sample

- What are typical values of t **under the H_0 hypothesis?**

Example

- Distribution of t under the H_0 hypothesis
- Vertical line = observed value of test statistics t
- Green = probability to observe under H_0 a lower value of t
- Blue = probability to observe under H_0 a larger value of t
- Here: Blue = 0.3% of total area



Conclusion: if H_0 (= no effect) is true, there is a **0.3% probability** to observe a value of t larger or equal to the one observed

→ **very unlikely, H_0 is probably not true and should be rejected**

Example

- **Study:** effect of fertilizer F3 on plant growth

- no-fertilizer: $h = 1.5m$
 - fertilizer on $n = 10$ samples:

$$X = \{1.47, 1.45, 1.31, 1.41, 1.47, 1.51, 1.55, 1.39, 1.44, 1.5\}$$

- **Question:** *does the treatment with fertilizer **enhance** plant growth?*

- **Hypothesis:**

- H_0 : no, $E(X) \leq h = 1.5m$
 - H_1 : yes $E(X) > h = 1.5m$

- Effect size:
 - size of random effect:
- $$\left. \begin{array}{l} \bar{x} - h \\ s/\sqrt{n} \end{array} \right\} t = \frac{\bar{x} - h}{s/\sqrt{n}} = -2.32$$

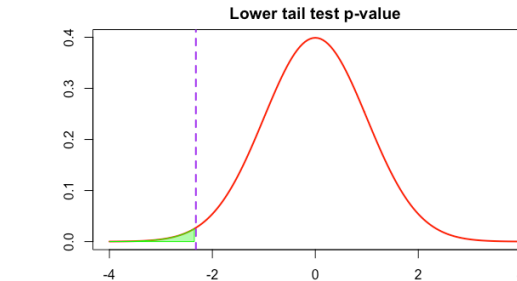
s = standard
deviation
of sample

- What are typical values of t **under the H_0 hypothesis**?

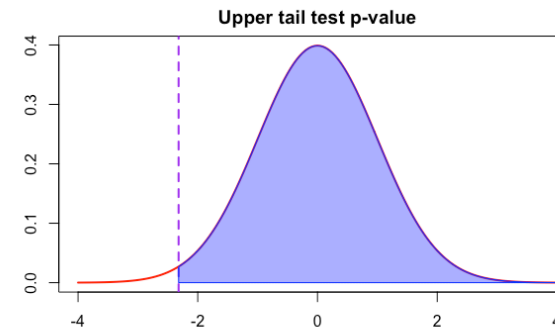
Example

$$t = -2.32$$

- Distribution of t under the H_0 hypothesis
- Vertical line = observed value of test statistics t
- Green = probability to observe under H_0 a lower value of t
- Blue = probability to observe under H_0 a larger value of t
- Here: Blue = 99% of total area



Pvalue = 0.010



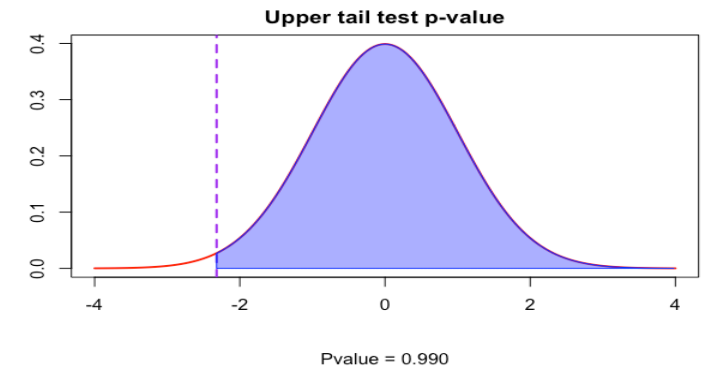
Pvalue = 0.990

Conclusion: if H_0 (= no effect) is true, there is a **99% probability** to observe a value of t larger or equal to the one observed

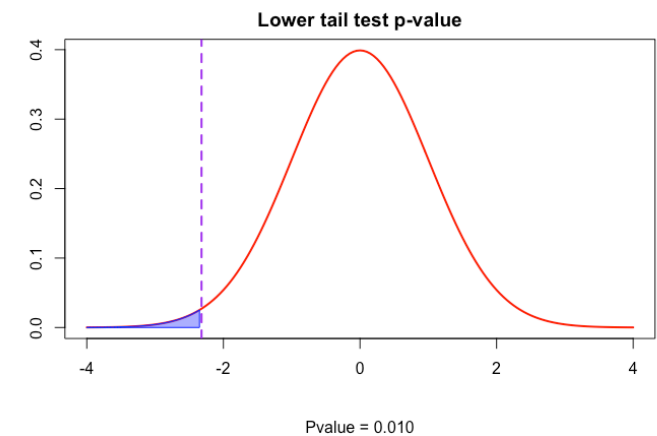
→ **very likely, H_0 cannot be rejected...**

What was the question again?

- Question 1:
*does the treatment with fertilizer **enhance** plant growth?*
(→ expected direction of effect is implicit: **“upper tail one-sided test”**)
H0: no! H1: yes!
- Question 2:
*does the treatment with fertilizer **reduce** plant growth?*
(→ expected direction of effect is implicit: **“lower tail one-sided test”**)
H0: no! H1: yes!



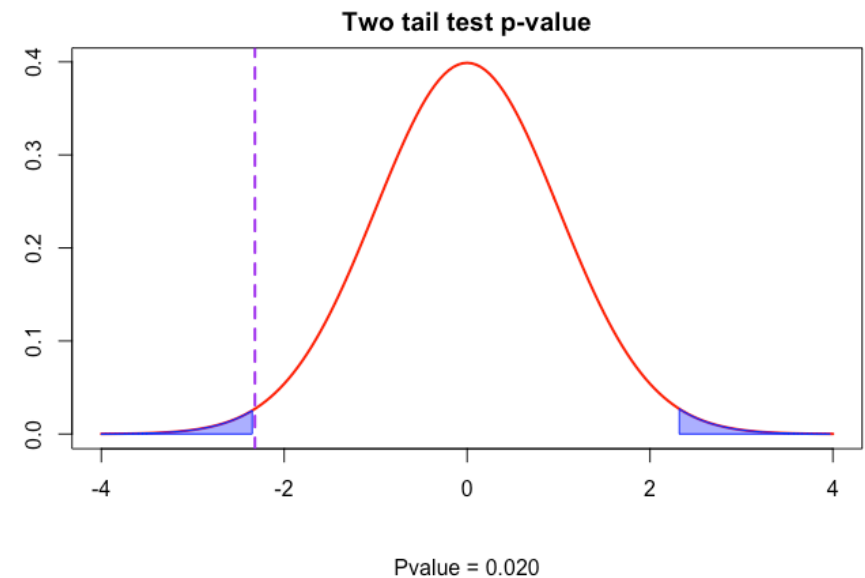
blue area = 99%: H0 cannot be rejected!



blue area = 1%: H0 very unlikely

What was the question again?

- Question 3:
*does the treatment with fertilizer **has any effect** on plant growth?*
(→ no direction implicit: **“two-sided test”**)
H0: no! H1: yes!



blue area = 2%: H0 very unlikely

P-value

The p-value is the **probability** of obtaining a

- larger (one-sided upper tail)
- smaller (one-sided lower tail)
- more extreme (two-sided or two tailed)

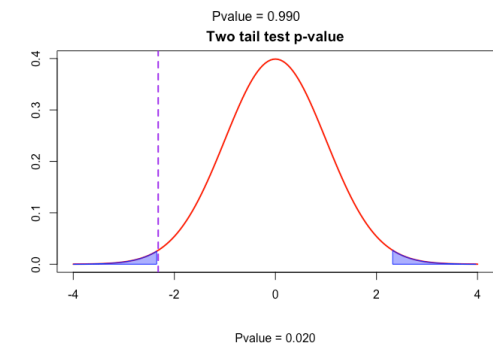
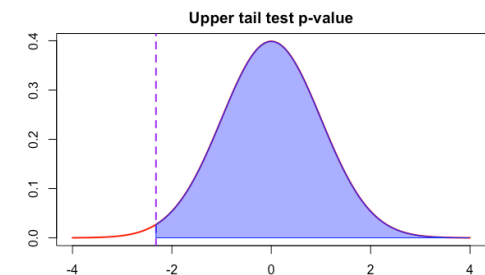
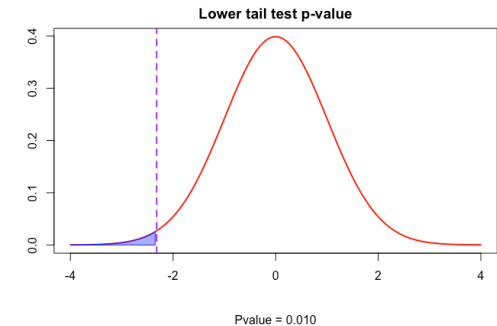
value of the test statistics **if H_0 is valid!**

The p-value represents the area under the H_0 curve

- above observed value (one-sided upper tail)
- below observed value (one-sided lower tail)
- more extreme than observed value
(two-sided or two tailed)

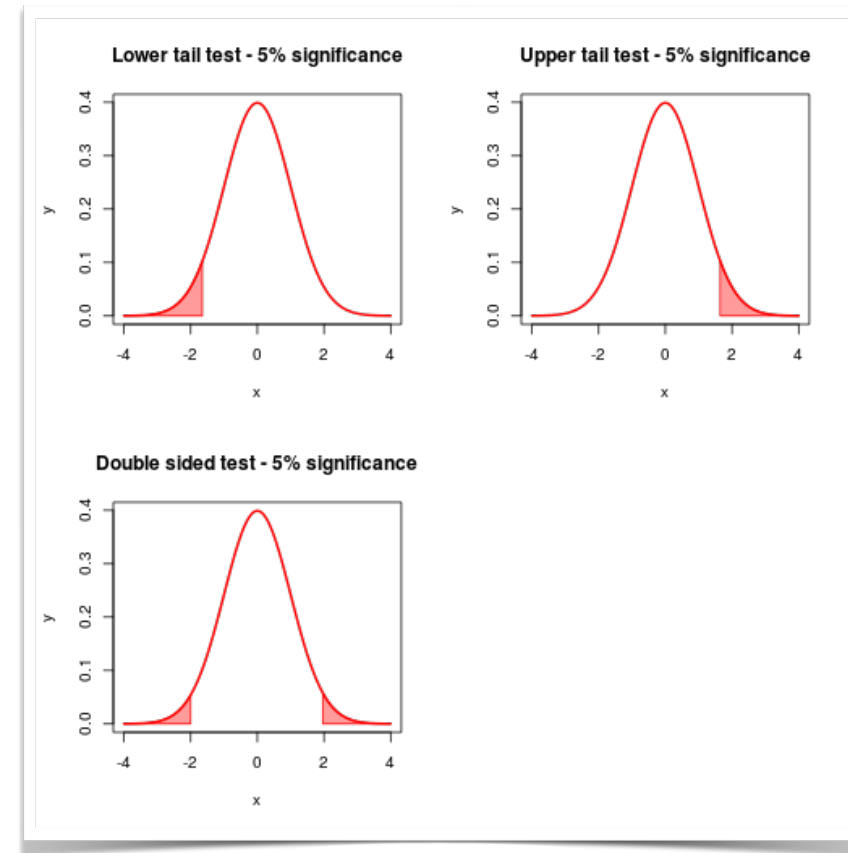
The probability of the two sided test is **twice** the smallest probability of the upper-tail or lower-tail test

$$p_{2sided} = 2\min(p_{lower-tail}, p_{upper-tail})$$



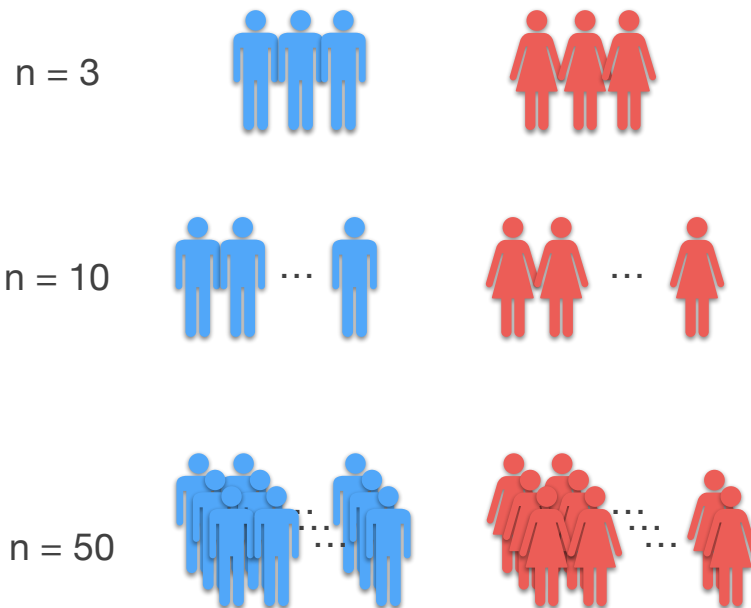
Significance

- When is a probability low, very low, or high?
- Define a **significance level α**
- $p < \alpha$:
 - H_0 hypothesis can be rejected
 - the observed effect is significant
 - H_1 is statistically proven
- $p > \alpha$:
 - effect is not sufficient to reject H_0
 - observed effect is compatible with statistical fluctuations
 - H_0 is not proven, maybe with a larger sample, the effect could become significant
- $\alpha = 0.05$ has become a standard value (but no golden rule!)



Effect size vs. significance

comparing mean weights



n	w.m	w.f	D	p	-log(p)
3	69,51	66,48	3,03	3,76E-02	1,425
10	70,08	66,98	3,10	1,42E-08	7,846
50	70,17	67,24	2,93	1,11E-24	23,957

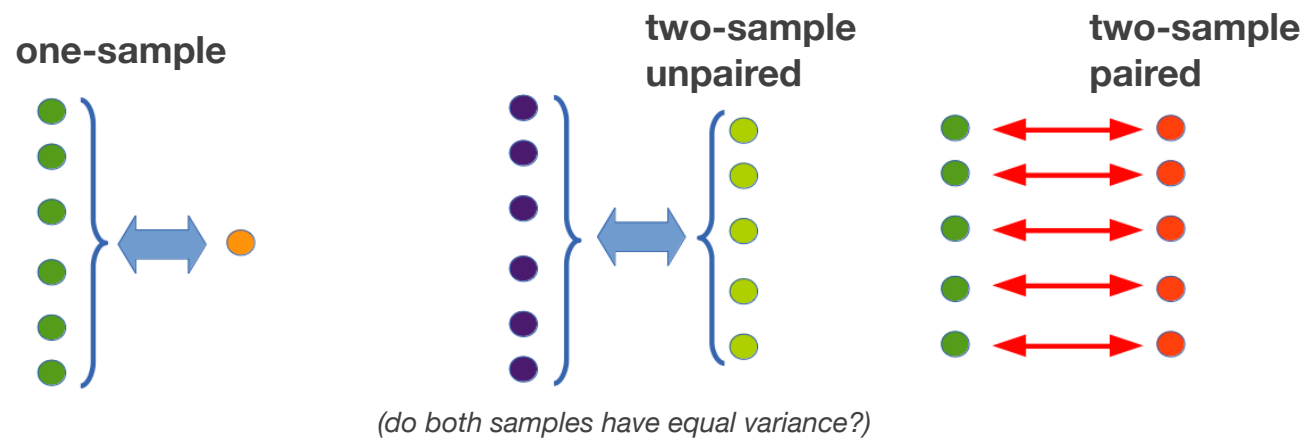
- A small effect size can become significant for large n
- A large effect size can be none-significant if n is low

7. Hypothesis testing

7.2 Testing the mean - t-tests

Test on mean values

- Hypothesis on mean values can be investigated using a ***t*-test**
- Family of tests with different version:
 - **one-sample test:** *is the mean body temperature 37.7 C?*
 - **two-sample test, unpaired:** *do men and women have different mean cholesterol levels?*
 - **two-sample test, paired:** *is there a change in cholesterol level after a one-month egg rich diet?*



t-test test statistics

Type	test statistics	degrees of freedom	note
one sample	$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$	n-1	
two-sample unpaired (same variance)	(Student t-test) $t = \frac{\bar{x}_1 - \bar{x}_2}{s_{12} \sqrt{1/n_1 + 1/n_2}}$	n ₁ +n ₂ -2	$s_{12} = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$
two-sample unpaired (diff. variance)	(Welch t-test) $t = \frac{\bar{x}_1 - \bar{x}_2}{s_{\bar{x}_1 - \bar{x}_2}}$	(*)	$s_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
two-sample paired	$t = \frac{\bar{x}_D - \mu}{s_D / \sqrt{n}}$	n-1	x_D = difference between pairs μ = expected difference

$$(*) \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)}$$

Distribution under H_0

- The test statistics of the t-tests under H_0 are distributed according to a **t-distribution** with the corresponding number of degrees of freedom
- for large sample sizes, the H_0 distribution is the standard normal distribution $\mathcal{N}(0,1)$

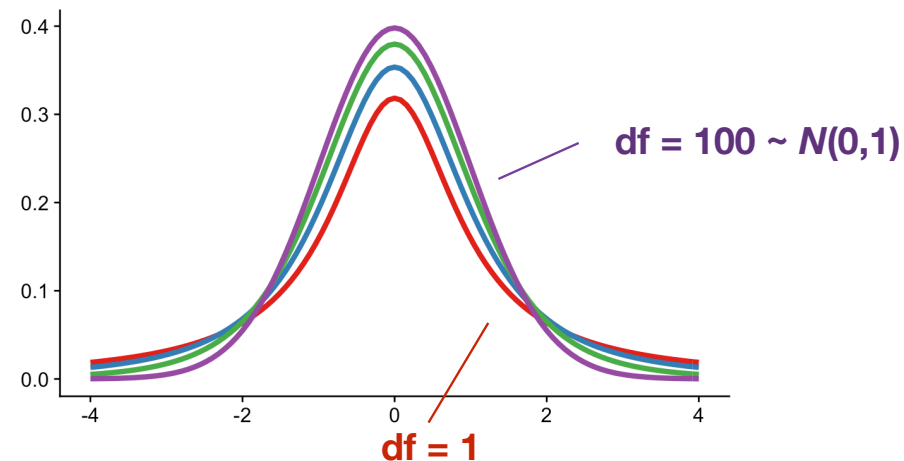
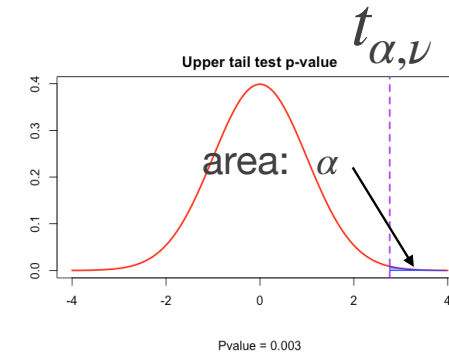


Table of critical values

two-sample t-test, unpaired, one-sided

ν \	.10	.05	.025	.01	.005	.001	.0005
1	3.078	6.314	12.706	31.821	63.657	318.31	636.62
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781

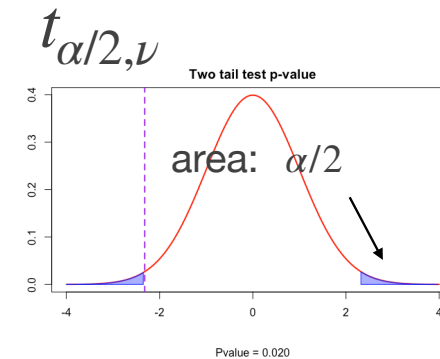


- Example (1-sample t-test)
 - $\alpha = 0.05$
 - $t = 2.01$
 - sample size $n = 8 \rightarrow \nu = n - 1 = 7$
- one-sided t-test
 - critical value $t_{0.05, 7} = 1.895$
 - $t > t_{0.05, 7}$: test is significant for $\alpha = 0.05$
 - **H_0 can be rejected: result is significant!**

Table of critical values

two-sample t-test, unpaired, two-sided

ν	.10	.05	.025	.01	.005	.001	.0005
1	3.078	6.314	12.706	31.821	63.657	318.31	636.62
2	1.886	2.920	4.303	6.965	9.925	22.326	31.598
3	1.638	2.353	3.182	4.541	5.841	10.213	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781



- Example (1-sample t-test)
 - $\alpha = 0.05$
 - $t = 2.01$
 - sample size $n = 8 \rightarrow \nu = n - 1 = 7$
- two-sided t-test
 - critical value $t_{0.025,7} = 2.365$
 - $t < t_{0.025,7}$: test is NOT significant for $\alpha = 0.05$
 - **H_0 cannot be rejected: test is NOT significant**

Running a t-test in R

two-sample unpaired, two-sided

t = test statistics
df = degrees of
freedom

confidence interval
differences of the
means

```
> t.test(weight.m,weight.f,var.equal=TRUE)

      Two Sample t-test
data:  weight.m and weight.f

t = 1.8265, df = 400, p-value = 0.06852

alternative hypothesis: true difference in
means is not equal to 0

95 percent confidence interval:
 -0.5669448 15.4259192

sample estimates:
mean of x mean of y
 181.9167  174.4872
```

Running a t-test in R

two-sample unpaired, one-sided

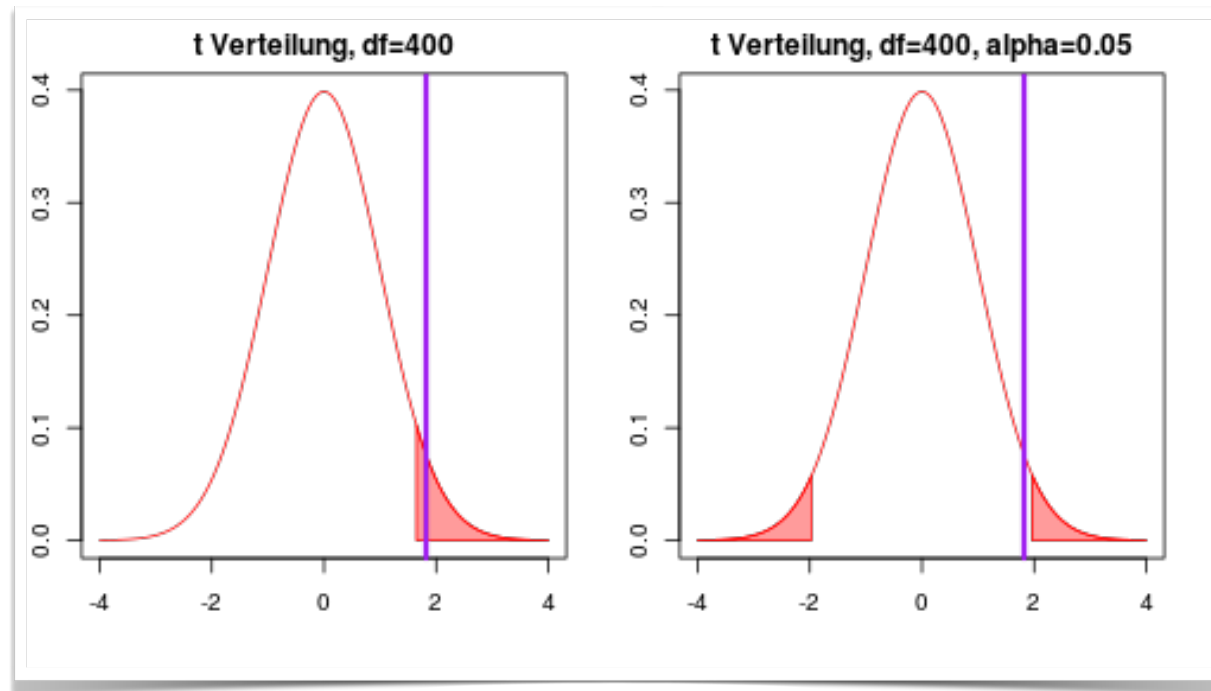
```
>t.test(weight.m,weight.f,alternative="greater",va  
r.equal=TRUE)
```

t = test statistics
df = degrees of
freedom

confidence interval
differences of the
means

```
Two Sample t-test  
data: weight.m and weight.f  
  
t = 1.8265, df = 400, p-value = 0.03426  
  
alternative hypothesis: true difference in means  
is greater than 0  
  
95 percent confidence interval:  
 0.723444      Inf  
sample estimates:  
mean of x mean of y  
181.9167 174.4872
```

Running a t-test in R



one-sided t-test
→ significant

two-sided t-test
→ non significant

Running a t-test in R

two-sample Welch unpaired, one-sided

t = test statistics
df = degrees of
freedom

confidence interval
differences of the
means

```
>t.test(weight.m,weight.f,alternative="greater")

      Welch Two Sample t-test
data:  weight.m and weight.f

t = 1.8453, df = 372.446, p-value = 0.0329

alternative hypothesis: true difference in means
is greater than 0

95 percent confidence interval:
 0.7903498      Inf

sample estimates:
mean of x mean of y
 181.9167  174.4872
```


Paired t-test

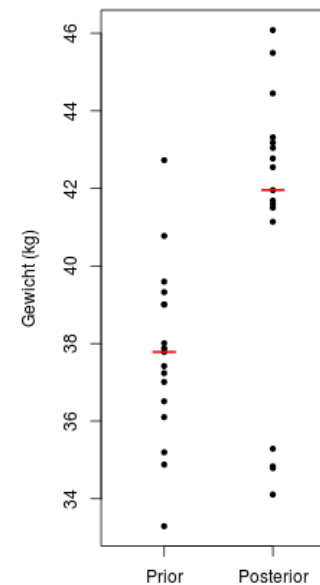
- 2 samples with equal number of elements
- each element of sample A can be associated to one element of sample B
 - patients before (A) and after (B) treatment
 - technical replicates

$$t = \frac{\bar{x}_D - \mu}{s_D / \sqrt{n}}$$

$\bar{x}_D$ = mean of differences

μ = expected difference

Treatment against anorexia
Weight before/after treatment



unpaired: $p = 5 \cdot 10^{-3}$

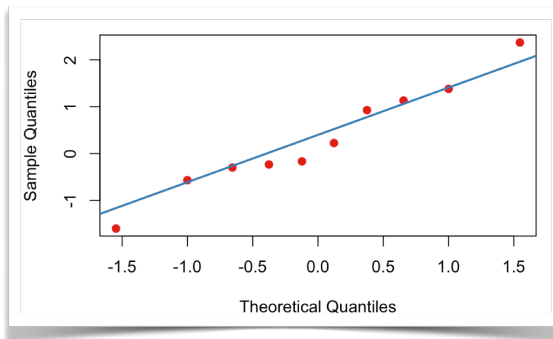
When can we apply t-test?

- There are several conditions that must be fulfilled to apply a t-test
- **Normality:** data must be (approximately) normally distributed
 - check using
 - QQ-plot
 - statistical tests: Shapiro-Wilks / Kolmogorov-Smirnov
 - if not, apply non-parametrical test
- **Variance** of samples must be equal
 - if so: Student t-test
 - if not: Welch t-test
- **Independance:** independent samples: values in one sample should not be influenced by those in the second sample

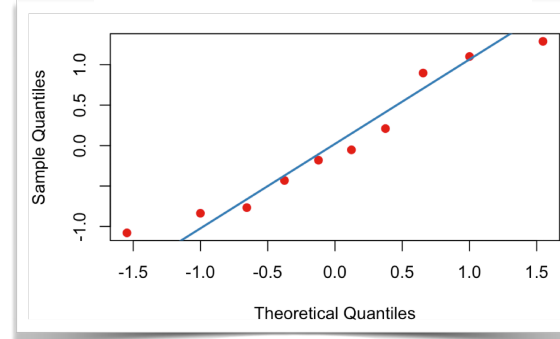
Checking normality

QQ-plot

normal distribution



non-normal distribution



Shapiro-Wilks test

H0: distribution is normally distributed

```
> n <- 10  
> x <- rnorm(n) ## Normal distrib.  
> shapiro.test(x)  
      Shapiro-Wilk normality test  
data:  x  
W = 0.977, p-value = 0.9471
```

H0 cannot be rejected
→ data normally distributed

```
> n <- 10  
> x <- rt(n,df=2) ## t-distribution  
> shapiro.test(x)  
      Shapiro-Wilk normality test  
data:  x  
W = 0.6966, p-value = 0.0008003
```

H0 should be rejected
→ data NOT normally distributed

Week 6 - stop and think

*First think about the questions on a sheet of paper;
then use your favorite AI to generate a small R code to illustrate these questions!*

- **Question 1:** A researcher wants to test whether a new medication reduces cholesterol levels. They measure cholesterol in 30 patients before treatment and again in the same 30 patients after 3 months of treatment. Which type of t-test should they use?
 - a) One-sample t-test
 - b) Two-sample unpaired t-test (Student)
 - c) Two-sample unpaired t-test (Welch)
 - d) Two-sample paired t-test
- **Question 2:** Explain the difference between a null hypothesis (H_0) and an alternative hypothesis (H_1).
- **Question 3:** Explain what a significance level $\alpha = 0.05$ means. If a study reports $p = 0.03$, what conclusion can you draw about the null hypothesis?
- **Question 4:** A p-value of 0.03 means that there is a 3% chance that the H_0 hypothesis is true. Right?Wrong?
- **Question 5:** Checking assumptions You want to compare cholesterol levels between two groups using a t-test. You create a QQ-plot for each group and notice that one group's points deviate substantially from the diagonal line. The Shapiro-Wilk test for this group gives $p = 0.003$. What should you do, and why?